

SWASTHYASYNC HEALTHCARE PLATFORM

Executive Product Overview, User Experience & Comprehensive Feature Guide

Platform Scope: All-in-One Healthcare Ecosystem for Citizens, Doctors & District Authorities | **Coverage:** Pan-India (36 States/UTs)

1. Executive Summary & Core Platform Vision

SwasthyaSync is a unified, India-wide digital health platform engineered to bridge the gap between citizens, frontline healthcare professionals, and district health governance. Built around a privacy-first, decentralized, and location-aware architecture, SwasthyaSync eliminates geographic fragmentation and brings seamless healthcare delivery to all 36 States and Union Territories.

Portal & Role	Primary User Personas	Key Workflow & Clinical Objectives
Citizen / Patient	Patients, Families, Rural Citizens, Chronic Disease Patients	Instant hospital discovery on live maps, OPD appointment booking, AI report simplification, health locker & dynamic ABHA QR profile.
Hospital & Doctor	Chief Medical Officers, Physicians, Surgeons, Nurses, Pharmacists	OPD live queue triage, digital e-prescription builder with generic alternatives, 360° EHR browser, emergency bed allocation.
District Administrator	District Health Officers (DHO), CMOH, Epidemiologists, Civil Surgeons	Real-time bed/ICU/oxygen surveillance across district hospitals, Integrated Disease Surveillance Programme (IDSP), outbreak alerts.

2. Citizen / Patient Portal Features

A. Real Hospital Discovery & Interactive Geospatial Map

- **Live GPS Radar:** Uses HTML5 device geolocation to calculate real Haversine distances to nearest healthcare facilities.
 - **Interactive Leaflet Map:** OpenStreetMap vector tiles with pulsing patient position pin and interactive medical markers.
 - **Dual Mode Fallback:** Switch between live GPS and Pan-India registered address with real-time Overpass API queries.
 - **Facility Details:** Live emergency casualty hotline, real-time bed count, available departments, and public/private badges.

B. AI Medical Report Simplifier & Comparative Viewer

- **Side-by-Side Dual-Pane Viewer:** View the original diagnostic laboratory report simultaneously with the AI-generated plain-language summary.
 - **Critical Parameter Highlighting:** Abnormal values flagged with red badges, normal parameters marked with green status badges.
 - **ABDM Digital Locker Integration:** Recommended one-click digital record saving into the patient's permanent encrypted locker.

C. Dynamic Health QR & ABHA Profile

- **Emergency QR Code:** Instant bedside scanning reveals blood group, known allergies, chronic conditions, and emergency contacts.
- **Longitudinal History:** Past appointments, digital prescriptions, vaccination logs, and downloadable health cards.

D. OPD Appointment Booking & Symptom Triage

- **Multi-Step Pan-India Booking:** Filter by State → District → City → Facility → Clinical Department → Doctor.
- **Interactive Symptom Triage:** Interactive symptom checker providing urgency grading (Normal, Moderate, Urgent) before booking.

3. Hospital Staff & Doctor Portal Features

- **Real-Time OPD Queue Command:** Live token status tracking (Waiting, In Consultation, Completed), instant patient calling with visual priority flags.
- **Digital Prescription (Rx) Builder:** Rapid prescription generator with automated generic drug alternatives, dosage scheduling, and instant pharmacy dispatch.
- **360° Electronic Health Record (EHR):** Searchable patient directory providing complete clinical history, past diagnoses, vitals, and lab summaries.
- **Facility Bed & Emergency Management:** Live monitoring of General, ICU, and Oxygen-supported bed occupancy.

4. District Health Administration Portal Features

- **Operations Command Center:** Comprehensive jurisdictional overview for District Health Officers (DHO/CMOH) covering all empaneled hospitals, PHCs, and Sub-Centres.
- **Integrated Disease Surveillance Programme (IDSP):** Real-time tracking of vector-borne, water-borne, and respiratory disease outbreaks with weekly trend analysis.
- **Emergency Broadcast System:** Issue high-priority epidemiological alerts, capacity warnings, and public health advisories across the district.
- **Infrastructure & Bed Tracking:** Aggregated utilization metrics, oxygen cylinder reserves, and critical care availability.
- **Governance & Health Reporting:** Automated maternal health, immunization, and institutional delivery analytics.

5. Universal Platform Capabilities

- **23-Language Instant Localization:** Seamless switching across English, Hindi, Marathi, Bengali, Telugu, Tamil, Kannada, Malayalam, Gujarati, Punjabi, Odia, Assamese, Urdu (with RTL layout), and 10 additional scheduled Indian languages via a single global header controller.
- **Location-Aware Identity System:** Centralized context ensures zero hardcoded geographic bias—supporting custom user registration and pre-configured multi-region test accounts (Delhi, Karnataka, Maharashtra).
- **DPDP Act 2023 & ABDM Privacy Compliance:** Strict patient consent controls, encrypted session storage, and role-based data isolation.